# Engineering Mathematics III

ENGE702

Lecture 4: Solving first order separable differential equations

# First order ODEs

In this lecture we will see how to find exact solutions to certain classes of differential equations.

The **order** of a differential equation is the level of the highest derivative in the equation. For example:

$cos(x)y'' + x^3y^5y' = 2$ has order 2 (second order),

$x^7y' + 2y^2 = \sin(y)$ has order 1 (first order),

$y^4 - 2y'' + \sin(y''''') = 2x$ has order 5 (fifth order),

etc.....

In this lecture we will see how to solve certain first order ODEs. Later we will see how to solve a small class of second order ODEs.

# Separable ODEs

Before we start on some more general classes, we will see how to solve the most basic first order equations.

**Question:** What is a function $y$ such that $y = y'$?

**Exercise** Based on this, come up with a function $y$ such that $ky = y'$ for a constant $k$.

**Exercise** Based on this, come up with a solution to the differential equation $y' + ky = 0$.

# Separable ODEs

The first class of differential equations we will see how to solve is the class of separable ODEs.

A first order ODE is **separable** if it can be written in the form

$$g(y)y' = f(x)$$

# Separable ODEs

We can (often) solve a separable ODE as follows:

Remember that $y'$ is another notation for $\frac{dy}{dx}$. Knowing this, we can rearrange our ODE to be

$$g(y)\,dy = f(x)\,dx$$

From there we can integrate both sides (if possible)

$$\int g(y)\,dy = \int f(x)\,dx$$

From here, we can find an explicit solution, ie $y = h(x)$, or an implicit solution $H(x, y) = 0$.

# Separable ODEs

**Example**: Find the general solution to the differential equation

$$y' + 2xy = 0$$

Note that we can rearrange this to be

$$\frac{dy}{dx} = -2xy$$
$$\frac{dy}{y} = -2x\ dx$$

From here we can integrate both sides and get

$$\ln(y) = -x^2 + c$$

Taking both sides to the power of $e$:

$$y = e^{-x^2+c}$$

or alternatively

$$y = Ce^{-x^2}$$

# Separable ODEs

Now we will look at another example, this time a more practical application.

**Example**: Newton's law of cooling states that the temperature $T(t)$ of a cooling object changes at a rate proportional to the difference between its temperature and the ambient temperature $T_A$, ie

$$\frac{dT}{dt} = k(T - T_A)$$

Suppose a cup of coffee has a temperature of $75^\circ$, and 10 minutes later it is $65^\circ$. The ambient temperature is $20^\circ$.

1. Find a formula for the temperature $T$ as a function of time $t$.
2. How long before the coffee cools to $50^\circ$?

# Separable ODEs

**Example**: Mixing problem:

A tank contains 1000 litres of water with 10kg of salt disolved into it. Brine (water with salt dissolved) containing 0.5kg of salt per litre is added to the tank at a rate of 10 litres per minute. This is mixed uniformly into the solution in the tank. The solution in the tank is also removed at a rate of 10 litres per minute.

Find a function that describes the concentration of salt in the tank at any time $t$.

# Text Book References

Chapter 1.3, 1.5